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Orbit Design and Control of Technology Validation Mission for Refractive Space Telescope in Formation Flying

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Many proposed formation flying missions seek to advance the state of the art in spacecraft science imaging by utilizing precision dual spacecraft formation flying to enable a "virtual" telescope (VT). Using precision alignment of two spacecraft very long focal lengths can be achieved by locating the optics on one spacecraft and the detector on the other. Furthermore, orders of magnitude improvement in science imaging over conventional single spacecraft instruments is possible with advances in precision formation flying. Precise astrometric alignment control is required for accurate inertial pointing of the VT. This work presents orbit design and control approaches for VT formation flying, with specific application to the recently proposed Virtual Telescope Demonstration Mission (VTDM) using cubesats. The characteristics of the mission orbit are derived from solutions to Hill's equations for the relevant mission orbit. A relative orbit control technique pertaining to the astrometric alignment problem is developed by utilizing the Constrained Model Predictive Control (CMPC) approach. This method enables design of formation flying control laws for a thruster assembly with limited magnitude. Precision formation flying is achieved by use of separate control laws for formation maintenance and precision astrometric alignment. The proposed control scheme compensates for inertial astrometric alignment errors as well as differences in orbital perturbations between the VTDM satellites. Numerical simulations are shown to validate the performance of the proposed relative orbit and alignment control laws for the VTDM. The results satisfy mission requirements and indicate the general applicability of the control algorithm to the VT formation flying architecture. Consequently, the proposed methods of orbit design and control can be applied to inertial astrometric alignment and formation keeping for the many proposed dual spacecraft precision formation flying missions.

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I. Introduction

THERE are many space telescopes that utilize the principle of a reflective telescope, such as Hubble Space Telescope, Spitzer Space Telescope, Planck Observatory, and James Webb Space Telescope. These space telescopes obtain images by reflective light with a single or combination of mirrors. Telescopes using reflective optics have the advantage of minimal chromatic aberration and require optical manufacturing precision for only a single reflective surface of each element. On the other hand, refractive telescopes bend light through an objective

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lens to converge at a focal point to make an image. This enables a much longer focal length telescope for higher resolution imaging, with particular use in high energy applications, such X-ray imaging. A longer focal length allows higher magnification of an image in a narrower field of view, which also minimizes chromatic aberration resulting in a sharper image. Telescopes using single spacecraft designs have strict mechanical alignment and stability requirements for large metering structures, making a longer focal length telescope a significant challenge. A potential solution uses two satellites instead, one with the refractive lens and the other with the detector. The separation between the two spacecraft can be used for an increased focal length telescope. The metering structure is replaced by precision formation flying using astrometric alignment control, thus enabling very long focal length telescopes. This architecture is known as a "virtual" telescope (VT).[5,17] Astrometric alignment refers to using dual-spacecraft precision formation flying to align the optics and detector on separated spacecraft along the line-of-sight from detector to a celestial target.

Many proposed missions use the VT astrometric alignment architecture. The Milli-Arc-Second Structure Imager (MASSIM) introduces a set of diffractive-refractive lenses on an optics satellite to focus X-rays onto detectors on a second satellite[1]. The proposed MASSIM requires two satellites in formation flying at a baseline separation of 1000 km in order to achieve X-ray imaging with milli-arc-second resolution. System requirements of two satellites in formation flying at 100 m baseline separation are also given for an application of X-ray solar flare imaging with 0.1 arcsec angular resolution[2]. Similarly to the MASSIM, the concept of the New Worlds Observer (NWO) requires two satellites in formation flying consisting of a telescope and an external starshade for measurement of extra-solar planets[3]. To enable a solar coronagraph, the PROBA-3 mission also provides space system consisting of two satellites separated by about 150 m[4]. In these applications, one satellite provides an external occulter to block the light from a primary source, providing a method to image exo-solar planets or solar corona, respectively using an optical instrument on the other satellite.

Precise satellite formation flying is necessary for required inertial alignment of an optics (or occulter) satellite with respect to a detector satellite for long baseline refracting space telescope [1,2], shadowing[3], or occulting missions[4], . The most challenging aspects of these missions are the relative navigation and control of satellites in formation flying. The inertial pointing requires very precise astrometric alignment control and estimation to make it possible for the "virtual" telescope to enable science imaging[5]. A study on covariance analysis of astrometric alignment estimation is performed for a space refracting telescope in formation flying[5]. Relative control for the astrometric inertial pointing also plays a key role in enabling the "virtual" telescope. Precise control and estimation requirements for relative spacecraft dynamics in a complex space environment provide a challenging system design. Initial flight applications will likely focus of shorter baselines, such as for heliophyiscs applications[2]. Thus, demonstration of a scalable formation flying architecture would facilitate extension of VT operation to larger baselines.

NASA Goddard Space Flight Center (GSFC) has proposed the Virtual Telescope Demonstration Mission (VTDM)[17] to develop and demonstrate new technologies required for a "virtual" telescope in a formation flying architecture that is scalable over a wide range of spacecraft separations. The VTDM offers a low-cost, small-scale route to demonstrate virtual telescope (e.g., a refractive space telescope) using two 6U cubesats, providing a necessary first step in the development of virtual telescope technology. The VTDM is composed of an Optics cubesat with a simple lens and a Detector cubesat with array of linear CCDs for demonstration of solar disk imaging[17]. A 3U version of this design has been proposed by a team at Yonsei University in consultation with the VTDM team at NASA GSFC. This VT design, as shown in Fig. 1, is called the Cubesat Astronomy by NASA and Yonsei using Virtual Telescope Alignment (CANYVAL). The VTDM tests five core technologies including, Guidance, Navigation and Control (GNC) system, astrometric alignment system, optics system, micro-propulsion system, and inter-satellite cross link system[17]. The GNC system for the VTDM requires control algorithm development for inertial astrometric alignment between two cubesats. Hence, the main objective of the current study is to investigate orbit design and control algorithm for this crucial technology. This algorithm is required to function with constrained control inputs since the orbit control will be accomplished using a thruster system with limited thrust magnitude. The Constrained Model Predictive Control (CMPC) approach is a well-known method to solve constrained quadratic regulation problems[6]. Hence, in the current study, the CMPC approach proposed in Bemporad et al.[7] is utilized to build an orbit controller for the CANYVAL detector satellite.

The current study has addressed multiple challenges, which are described herein. First, a proper orbit design to meet the mission objectives is introduced by using the Hill's dynamics and solutions. Second, a relative orbit control algorithm for the astrometric alignment phase is developed via the CMPC technique. Third, separate orbit maneuver strategy is proposed for regulating and formation keeping controls. Fourth, numerical simulations are performed to give insights into the developed algorithm and to evaluate the compatibility of the controller with severe orbital perturbations. This paper is organized as follows. Section 2 outlines representative relative orbit and attitude

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requirements for the CANYVAL mission that will be used for evaluation of the proposed control approach. Section 3 briefly introduces the Hill's relative orbital dynamics and presents analytical solutions to the Hill's equations. By utilizing the solutions to the Hill's dynamics, an orbit of the CANYVAL is designed as an example. Section 4 introduces the CMPC technique to build an orbit control algorithm for the astrometric alignment mission in formation flying while handling the constraint on thrust magnitude of the CANYVAL cubesat. This section also includes an explanation of orbit maneuver and formation keeping strategies. Section 5 presents orbital perturbation analysis and numerical simulations showing that the desired astrometric alignment and formation keeping for CANYVAL can be successfully conducted with reasonable limits on thrust level.

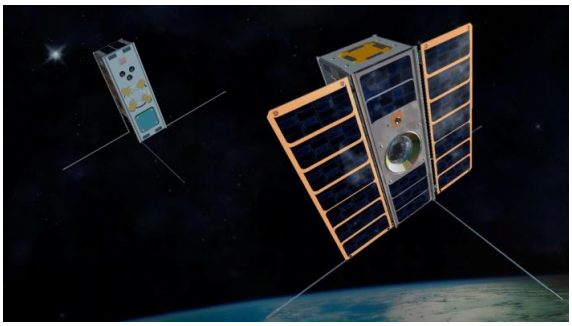


Figure 1. The CANYVAL Cubesats in Astrometric Alignment

II. Orbit and Attitude Requirements for the Astrometric Alignment

During the VT operation the two cubesats are maneuvered to be aligned with the Sun in order to verify the astrometric alignment performance of the VT axis to a celestial target. The astrometric alignment with the Sun is held for a few minutes to achieve steady-state. A lens in the Optics cubesat focuses sunlight on a CCD mounted to the Detector cubesat. This CCD should be located entirely in the umbra of the Optics cubesat to avoid stray light for obtaining a precise measurement of the solar alignment. However, this constraint requires very close separation of the two cubesats when considering the shortest dimension (10 cm) of a standard 3U cubesat and the size of the CCD array (7cm). So, the CANYVAL optics cubesat will use deployable solar panels to provide additional power (as shown in Fig. 1) and to increase the umbral length, providing ample room to avoid collision of the two cubesats during proximity operations before and after VT alignment periods. With the panels deployed the minimum side length of the Optics cubesat is increased to about 30 cm and consequently the length of umbra cast by the Optics cubesat is about 34 m. Based on the size (~7 cm) of the CCD on the Detector cubesat, a 20 m relative distance between the two satellites during astrometric alignment would provide ample margin for maintaining the CCD in the umbra while also providing a safe spacecraft separation.

Representative relative orbit and attitude requirements for the astrometric alignment for CANYVAL are summarized in this section. Attitude axes are defined such as: roll-axis is along the VT astrometric alignment axis (desired sun direction) and the pitch and yaw axes satisfy a right-handed coordinate system with respect to the roll-axis. In addition to the Hill's frame, used for proximity operation analysis in other section of this paper, a relative orbit frame is defined for VT astrometric alignment control. The relative orbit axes are defined as a right-handed system with radial, in-track, and cross-track components with origin at the Detector cubesat. The radial direction is along the line-of-sight from the Dececter to the Optics cubesat (along the sun direction during VT solar pointing operation). The in-track and out-of-track directions satisfy a right-handed coordinate system with respect to the radial direction. Then, a desired sun direction (i.e., the astrometric alignment direction) at a given time is the reference axis for both the attitude and the relative position. The relative distance, the diameter of image in the Detector cubesat, and the diameter of CCD on the Detector cubesat are assumed to be 20 m, 5 cm, and 9.2 cm, respectively. These design parameters are used as a basis to establish requirements for relative orbit and attitude control for CANYVAL. The basic criterion for the astrometric alignment is to align the Optics cubesat with the line-of-sight from the detector to the Sun (or other celestial target). The solar image focused by the lens on the Optics cubesat should be maintained on the Detector cubesat's CCD array (which is also within the umbra from the Optics cubesat). Table 1 shows the orbit and attitude requirements for the CANYVAL mission (all specifications are 3 sigma). The per-axis control requirements listed for each cubesat divide up the control architecture, with the Optics cubesat controlling in-track and out-of-track position and the Detector controlling the radial position. The change in the size of image on the Detector cubesat is not sensitive to the relative range between two satellites. So, about 10

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cm of range knowledge and about 50 cm range control would be accurate enough. Since the Detector cubesat receives the sun light passing through the lens on the Optics cubesat the astrometric alignment is not particularly sensitive to the Detector cubesat attitude error for the purpose of the CANYVAL alignment demonstration. So, for the Detector cubesat, about 5° of attitude knowledge and about 5° attitude control would be sufficient. To achieve the desired astrometric alignment, the attitude accuracy of the Optics cubesat is very important because the location of the solar disk image at the detector, focused by the lens on the Optics cubesat, is strongly affected by the attitude of the Optics cubesat. The relative translation accuracy of the Optics cubesat with respect to the desired astrometric alignment is also very important, because the Optics should be correctly located along the desired astrometric alignment axis (i.e. toward the Sun in this case). Table 2 shows the astrometric alignment budget (all specifications are 3 sigma) for a 20 m cubesat separation during VT alignment operations. The predicted translation error is the root square summation of all effective translation errors. The allowable translation error is determined by the size (9.2 cm diameter) of each CCD attached on the Detector cubesat and a 5 cm diameter solar disk image size.

Table 1 Orbit and attitude requirements for astrometric alignment

Orbit	Attitude	Optics Sat	Detector Sat
knowledge		≤ 2.0 cm	≤ 10 cm
radial control		none	≤ 50 cm
in-track control		≤ 2.0 cm	none
out-of-track control		≤ 2.0 cm	none
	knowledge	roll, pitch, yaw $\leq 0.06^o$	pitch, yaw $\leq 5^o$
	control	roll, pitch, yaw $\leq 0.06^o$	pitch, yaw $\leq 5^o$
	stability	none	none

Table 2 The astrometric alignment budget

	Knowledge	Control	Total	Effective Translation Error
attitude	0.06^o	0.06^o	0.085^o	2.8 cm
relative alignment	2.0 cm	2.0 cm		2.8 cm
relative range	10.0 cm	50.0 cm	51.0 cm	0.13 cm
relative alignment				2.0 cm
total predicted translation error				4.2 cm
allowable translation error				4.2 cm

III. Orbit Design for the Astrometric alignment

A. The Hill's equations

Linearized equations of relative motion of satellites are well known in the open literature. Excluding any perturbations, their form is restricted to two-body dynamics. In addition, they usually assume that the distance between two satellites is very small compared to the distance between the satellite and the Earth. The relative motion dynamics in two-body circular reference orbit known as the Hill's equations of motion[8] will be used for this study. A reference frame that appropriately expresses relative motion is needed. A local vertical local horizontal (LVLH) reference frame is chosen as one of the rotating reference frames. The first axis vector $\hat{x}$ is defined as the unit vector along the radial direction of the satellite. The third axis vector $\hat{z}$ is defined as the unit vector normal to the reference orbit plane. The second axis vector $\hat{y}$ is determined such that it completes the right-handed coordinate system. Relative position vector $\vec{r}$ is defined as difference of position between the two satellites. Relative position, velocity vectors and relative state vector can be expressed as following manner.

$$\vec{r} = [x\ y\ z]^T,\ \dot{\vec{v}} = [\dot{x}\ \dot{y}\ \dot{z}]^T,\ \vec{X} = [\vec{r}^T\ \vec{v}^T]^T = [x\ y\ z\ \dot{x}\ \dot{y}\ \dot{z}]^T \quad (1)$$

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After considering all the factors, the Hill's equations of motion can be expressed as follows[8].

$$\begin{bmatrix} \ddot{x} \\ \ddot{y} \\ \ddot{z} \end{bmatrix} = n^2 \begin{bmatrix} 3 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} + n \begin{bmatrix} 0 & 2 & 0 \\ -2 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{z} \end{bmatrix} \tag{2}$$

where n is mean motion of the reference satellite that is the Detector satellite.

B. Solutions to the Hill's equations

According to the Hill's equations, the solution of the relative motion in Eq. (2) can be expressed as follows [9] :

$$\begin{aligned}
x(t) &= \frac{\dot{x}_0}{n}\sin[n(t-t_0)] - \left(3x_0 + \frac{2\dot{y}_0}{n}\right)\cos[n(t-t_0)] + \left(4x_0 + \frac{2\dot{y}_0}{n}\right) \\
y(t) &= \left(6x_0 + \frac{4\dot{y}_0}{n}\right)\sin[n(t-t_0)] + \frac{2\dot{x}_0}{n}\cos[n(t-t_0)] - (6nx_0 + 3\dot{y}_0)(t-t_0) + \left(y_0 - \frac{2\dot{x}_0}{n}\right) \\
z(t) &= z_0\cos[n(t-t_0)] + \frac{\dot{z}_0}{n}\sin[n(t-t_0)] \\
\dot{x}(t) &= \dot{x}_0\cos[n(t-t_0)] + (3nx_0 + 2\dot{y}_0)\sin[n(t-t_0)] \\
\dot{y}(t) &= (6nx_0 + 4\dot{y}_0)\cos[n(t-t_0)] - 2\dot{x}_0\sin[n(t-t_0)] - (6nx_0 + 3\dot{y}_0) \\
\dot{z}(t) &= -z_0 n\sin[n(t-t_0)] + \dot{z}_0\cos[n(t-t_0)]
\end{aligned} \tag{3}$$

As we define t_0 as the instant at the ascending node of the reference orbit, $n(t-t_0)$ can be expressed as angle of latitude $\theta = \nu + \omega$ ($\theta(t) = nt, \theta(t_0) = 0$). Using the trigonometric relation of Eq. (4) and the condition for periodic motion Eq. (5) (i.e., bounded orbit), the Eq. (3) can be reduced to Eq. (6). The condition for periodic motion is just the requirement that the semi-major axes of the Optics cubesat and the Detector cubesat must be equal [10]. Then, the period of the relative orbit of the Optics cubesat is same as the orbital period of the Detector cubesat that is reference satellite.

$$a\sin(k) + b\cos(k) = \sqrt{a^2+b^2}\,\sin(k+\delta_1) = \sqrt{a^2+b^2}\,\cos(k+\delta_2) \tag{4}$$

$$\text{where } \delta_1 = \tan^{-1}\left(\frac{b}{a}\right),\ \delta_2 = \tan^{-1}\left(\frac{-a}{b}\right)$$

$$2x_0 + \frac{\dot{y}_0}{n} = 0 \tag{5}$$

$$\begin{aligned}
x(t) &= \frac{\dot{x}_0}{n}\sin\theta + x_0\cos\theta \quad &= c_1\sin(\theta+\alpha) \\
y(t) &= (y_0 - \frac{2\dot{x}_0}{n}) - 2\dot{x}_0\sin\theta + \frac{2\dot{x}_0}{n}\cos\theta = c_2 + 2c_1\cos(\theta+\alpha) \\
z(t) &= z_0\cos\theta + \frac{\dot{z}_0}{n}\sin\theta \quad &= c_3\sin(\theta+\beta) \\
\dot{x}(t) &= \dot{x}_0\cos\theta - nx_0\sin\theta \quad &= nc_1\cos(\theta+\alpha) \\
\dot{y}(t) &= -2nx_0\cos\theta - 2\dot{x}_0\sin\theta = -2nc_1\sin(\theta+\alpha) \\
\dot{z}(t) &= -z_0 n\sin\theta + \dot{z}_0\cos\theta \quad &= nc_3\cos(\theta+\beta)
\end{aligned} \tag{6}$$

$$\text{where } c_1 = \sqrt{x_0^2 + (\dot{x}_0/n)^2},\ c_2 = (y_0 - 2\dot{x}_0/n),\ c_3 = \sqrt{z_0^2 + (\dot{z}_0/n)^2}$$

$$\alpha = \tan^{-1}\left(\frac{nx_0}{\dot{x}_0}\right),\ \beta = \tan^{-1}\left(\frac{nz_0}{\dot{z}_0}\right)$$

When $\alpha = \beta$, Eqs. (6) yield a representation of a 3-dimensional ellipse that is generated by a section of 3-dimensional cylinder [10]. This ellipse is centered on the y-axis at (0, c_2, 0). We prefer the center of relative motion of

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the Optics cubesat that is located at the reference satellite, the Detector cubesat. This means that a required relative orbit should satisfy $c_2 = 0$ (*i.e.*, $y_0 = 2\dot{x}_0 / n$) as well as $\alpha = \beta$. Then, we obtain the following equations.

$$\begin{aligned} x(t) &= c_1 \sin(\theta + \alpha) \\ y(t) &= 2c_1 \cos(\theta + \alpha) \\ z(t) &= c_3 \sin(\theta + \alpha) \\ \dot{x}(t) &= nc_1 \cos(\theta + \alpha) \\ \dot{y}(t) &= -2nc_1 \sin(\theta + \alpha) \\ \dot{z}(t) &= nc_3 \cos(\theta + \alpha) \end{aligned} \tag{7}$$

where $c_1 = \sqrt{x_0^2 + y_0^2 / 4}$, $c_3 = \sqrt{z_0^2 + (\dot{z}_0 / n)^2}$, $\alpha = \tan^{-1}\left(\frac{2x_0}{y_0}\right) = \tan^{-1}\left(\frac{nz_0}{\dot{z}_0}\right)$

Equation (7) gives a representation of a 3-dimensional ellipse that is centered at (0, 0, 0). A relative periodic orbit satisfying Eq. (7) has its projection in the *x*-*y* plane of a 2:1 with the long axis as *y*-axis. When $c_3 = \sqrt{3}c_1$, Eq. (7) yields the relationship showing the circular relative orbit as follows.

$$x^2 + y^2 + z^2 = 4c_1^2 \tag{8}$$

The relationship keeps $z = \sqrt{3}x$ in the above equation. This means the circular relative orbit has an inclined angle of 60° with respect to *x*-*y* plane or an inclined angle of 30° with respect to the horizontal plane (*y*-*z* plane)[10].

C. Analysis of relative orbit design

Now by using the solutions to the Hill's equations, we can design a bounded relative orbit. In other words, we can choose appropriate initial position and velocity of the Optics and Detector cubesats in the Hill coordinate according to a desired bounded relative orbit. We adopt four conditions such as i) no cross-track component in the relative orbit, ii) a bounded relative orbit, iii) a relative motion centered at the reference satellite and iv) a limited minimum distance for collision avoidance. To save fuel consumption for the astrometric alignment keeping of the VT, the relative orbital plane in the Earth-Centered Inertial coordinate (ECI) should be parallel to solar light direction from the Sun. Then, a cross-track maneuver (z-axis in the Hill coordinate) is not necessary. This orbit has the solar light direction to be always in x-y plane in the Hill coordinate or in-plane in the ECI coordinate. This means that the Optics cubesat should always be in x-y plane to keep a desired inertial alignment between [the Detector cubesat – the Optics cubesat – the Sun]. This constraint yields an elliptic relative orbit, because a circular relative orbit (Eq. 8) requires an inclined angle of 60° with respect to *x*-*y* plane. This first condition will be relaxed if necessary, because this condition yields a strict condition of orbital inclination of the cubesats. However, in the current study, this condition is used as only an example. The condition for a bounded relative motion of the Optics cubesat around the Detector cubesat is just the requirement that the semi-major axes of the Optics and the Detector cubesats must be equal. This condition is enforced in Eq. (5). The solar light direction is almost constant in the ECI coordinate for a short duration (one-orbit period (a few hours) of satellite with respect to the Earth), because the Earth moves around the Sun for a one-year period. The solar light direction viewed from the reference cubesat (the Detector in Hill coordinate) is changed 360°/orbit (mean rate-1) according as the satellite moves around the Earth. The Optics cubesat viewed from the reference cubesat (the Detector in Hill coordinate) is also changed 360°/orbit (mean rate-2). Note that these values are *mean* changes of solar light direction and the Optics cubesat direction viewed from the Detector cubesat. Without maneuvering of satellite, the perfect condition to have an astrometric alignment of between [Detector cubesat – Optics cubesat – the Sun] is that an *instantaneous* rate-1 is the same as an *instantaneous* rate-2. A circular relative orbit may yield this perfect condition. However, there is no perfect condition because a circular relative orbit should have z-component that violates the first condition. So, a preferred relative orbit is centered at the reference satellite in order to have a similar *instantaneous* rate-1 to an *instantaneous* rate-2 as close as possible. This means that a required relative orbit should satisfy $c_2 = 0$ (*i.e.*, $y_0 = 2\dot{x}_0 / n$) as well as $\alpha = \beta$. A minimum distance of the Optics cubesat from the Detector cubesat should be limited to avoid a possible collision. A relative periodic orbit satisfying Eq. (7) has its projection in the *x*-*y* plane of a 2:1 with the long axis as *y*-axis. Thus, c_1 value yields a minimum distance between two satellites. Using these conditions explained above, we can design a reference orbit and a relative orbit. The position and velocity components in the Hill coordinate for the cubesats are following: Epoch time is assumed as [2014 year, 06 month, 01 day, 12 hour, 0 minute, 0 second], orbital elements of the Detector (reference) cubesat with about 600km altitude are [*a* (semimajor axis), *i* (inclination), Ω (longitude of ascending node), ω (argument of perigee), *e* (eccentricity), *M* (mean anomaly)] =

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[6978.000 km, 27.172°, 17.2° (0.3 rad), 0, 0, 85.9°(1.5 rad)], orbital elements of the Optics cubesat corresponding to the relative orbit are [a, i, Ω, ω, e, M]= [6978.000, 27.172° , 17.2° (0.3rad) , 224.534°, 0.000002866, -138.590°], the corresponding relative states of the Optics satellite in Hill coordinate are $[x_0, y_0, z_0, \dot{x}_0, \dot{y}_0, \dot{z}_0]$ = [0.015000 km, -0.026457 km, 0 km, -0.000014328 km/s, -0.000032493 km/s, 0 km/s].

The CANYVAL mission consists of two cubesats. The Detector cubesat is a reference satellite that is in a near-circular orbit around the Earth. The Optics cubesat has relative motions with respect to the Detector satellite. Based on the approach explained above, Figure 2 shows a designed relative orbit of the Optics cubesat with respect to the Detector in the Hill coordinate. The minimum and maximum distances between the two cubesats are 20 m and 40 m, respectively. The sun direction is defined as the direction of line of sight from Detector cubesat to the Sun. The sun direction is constantly and continuously changed in the Hill coordinate because the Detector (reference) cubesat in a near circular orbit moves around the Earth, while the sun direction is constant in the ECI coordinate for a relatively short duration. In Figure 2, the little circles mark relative positions of the Optics cubesat at four specific times equally spaced intervals (3 minutes). The long arrows indicate the sun directions at the same specific times for the specified relative positions. It can be figured out that the relative positions have different relative velocity because the relative orbit is ellipse, while the sun directions have constant velocity in the Hill coordinate. The relative velocity of the Optics cubesat is repeatedly changed in a pattern with (slow phase –fast phase – slow phase –fast phase) for one period in the Hill coordinate. Consequently, the relative positions of the Optics cubesat meet four times the sun direction that is constantly moved around the Detector cubesat. Figure 3 shows the angle profile between the Optics cubesat and the sun direction. As expected, the angle becomes zero four times when the Optics cubesat is aligned with the sun direction from the Detector cubesat. When the Optics cubesat has a positive x-component, an astrometric alignment of [the Detector cubesat – the Optics cubesat – the Sun] is possible. Thus, there are two desired alignments for the astrometric alignment for one orbital period. We can utilize these two chances to make a desired astrometric alignment.

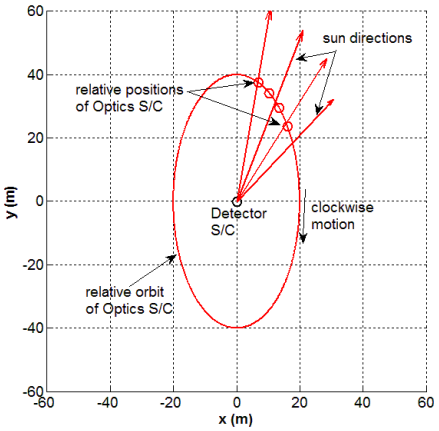


Figure 2. A designed relative orbit for the CANYVAL

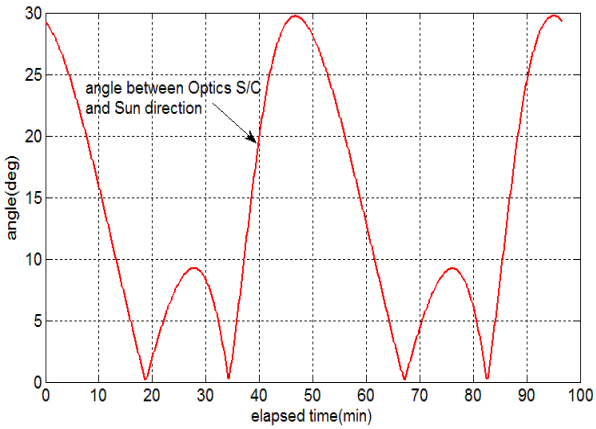


Figure 3. Angle profile between the Optics cubesat and the Sun direction in the relative orbit described in Figure 2.

IV. Controller Design for the Inertial Alignment

A. Constrained Model Predictive Control for the VT

The desired astrometric alignment of the VT requires the following position relationship between the Detector cubesat, the Optics cubesat and the Sun in the ECI coordinate.

$$\vec{r}_{\text{optics,derired}} = \vec{r}_{\text{detector}} + l\frac{(\vec{r}_{\text{sun}} - \vec{r}_{\text{detector}})}{\|\vec{r}_{\text{sun}} - \vec{r}_{\text{detector}}\|} \tag{9}$$

where $\vec{r}_{\text{optics,desired}}$ is a desired position vector of the Optics cubesat for the astrometric alignment, $\vec{r}_{\text{detector}}$ is position vector of the Detector cubesat, $\vec{r}_{\text{sun}}$ is position vector of the Sun and l is a desired distance between the two satellites. This relationship is the same as that obtained for the PROBA-3 [11] that also needs an inertial alignment for solar coronagraph. By differentiating Eq. (9) with respect to time, the desired velocity vector of the Optics cubesat in ECI coordinate can be obtained.

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$$\dot{\vec{r}}_{\text{optics,desired}} = \dot{\vec{r}}_{\text{detector}} + l\frac{dt}{d}\left(\frac{(\vec{r}_{\text{sun}}-\vec{r}_{\text{detector}})}{\|\vec{r}_{\text{sun}}-\vec{r}_{\text{detector}}\|}\right) \tag{10}$$

Let's define $\vec{r}_{\text{sun}} - \vec{r}_{\text{detector}} \equiv \vec{r}_{\text{sd}}$ then

$$\frac{dt}{d}\left(\frac{(\vec{r}_{\text{sun}}-\vec{r}_{\text{detector}})}{\|\vec{r}_{\text{sun}}-\vec{r}_{\text{detector}}\|}\right) = \frac{dt}{d}\left(\frac{\vec{r}_{\text{sd}}}{\|\vec{r}_{\text{sd}}\|}\right) = \frac{dt}{d}\frac{\vec{r}_{\text{sd}}}{r_{\text{sd}}} = \frac{\dot{\vec{r}}_{\text{sd}}}{r_{\text{sd}}} - \frac{\vec{r}_{\text{sd}}(\vec{r}_{\text{sd}}\cdot\dot{\hat{r}}_{\text{sd}})}{r_{\text{sd}}^2}$$

where $\hat{r}_{\text{sd}}$ is the unit vector of $\vec{r}_{\text{sd}}$. The relationship between the current velocity and the desired velocity at the current time of the Optics cubesat becomes as follows:

$$\dot{\vec{r}}_{\text{optics}} - \dot{\vec{r}}_{\text{optics,desired}} = \dot{\vec{r}}_{\text{optics}} - \dot{\vec{r}}_{\text{detector}} - l\frac{dt}{d}\left(\frac{(\vec{r}_{\text{sun}}-\vec{r}_{\text{detector}})}{\|\vec{r}_{\text{sun}}-\vec{r}_{\text{detector}}\|}\right) \tag{11}$$

By differentiating Eq. (11), the equations of the Optics cubesat with respect to its desired orbit in ECI coordinate can be obtained as follows :

$$\begin{aligned}\ddot{\vec{r}}_{\text{optics}} - \ddot{\vec{r}}_{\text{optics,desired}} &= \ddot{\vec{r}}_{\text{optics}} - \ddot{\vec{r}}_{\text{detector}} - l\frac{dt^2}{d^2}\left(\frac{(\vec{r}_{\text{sun}}-\vec{r}_{\text{detector}})}{\|\vec{r}_{\text{sun}}-\vec{r}_{\text{detector}}\|}\right) + \vec{u} \\ &= f(\vec{r}_{\text{optics}}) - f(\vec{r}_{\text{detector}}) - l\frac{dt^2}{d^2}\left(\frac{(\vec{r}_{\text{sun}}-\vec{r}_{\text{detector}})}{\|\vec{r}_{\text{sun}}-\vec{r}_{\text{detector}}\|}\right) + \vec{u}\end{aligned} \tag{12}$$

where $f(\vec{r})$ is natural accelerations including orbital perturbations. The right hand side of Eq. (12) should be zero in order to keep the desired astrometric alignment. Then, the control becomes

$$\vec{u} = f(\vec{r}_{\text{detector}}) - f(\vec{r}_{\text{optics}}) + l\frac{dt^2}{d^2}\left(\frac{(\vec{r}_{\text{sun}}-\vec{r}_{\text{detector}})}{\|\vec{r}_{\text{sun}}-\vec{r}_{\text{detector}}\|}\right) \tag{13}$$

The last term in the right side of Eq. (13) is so small that it is ignored, and control should mitigates the errors in position and velocity of the Optics cubesat with respect to the desired position and velocity of the Optics in order to keep the astrometric alignment. Thus the control finally becomes the same form as that derived for the PROBA-3 mission [11]:

$$\vec{u} = f(\vec{r}_{\text{detector}}) - f(\vec{r}_{\text{optics}}) + \vec{u}_e \tag{14}$$

where $\vec{u}=\begin{bmatrix}u_I & u_J & u_K\end{bmatrix}^T$ and $\vec{u}_e$ is the term to mitigate the errors in position and velocity of the Optics satellite.

The error equation has the following form [11]:

$$\dot{\vec{X}}_e = A\vec{X}_e + B\vec{u}_e \tag{15}$$

$$\text{where } A = \begin{pmatrix} 0_3 & I_3 \\ \frac{-\mu}{\|\vec{r}_{\text{detector}}\|^3}I_3 & 0_3 \end{pmatrix},\ B = \begin{pmatrix} 0_3 \\ I_3 \end{pmatrix}$$

$$\vec{X}_e = \begin{pmatrix} \vec{r}_{\text{optics}} - \vec{r}_{\text{optics,derired}} \\ \dot{\vec{r}}_{\text{optics}} - \dot{\vec{r}}_{\text{optics,desired}} \end{pmatrix}$$

The $\vec{u}_e$ is computed by using a linear control technique because the error equation of Eq. (15) is linear. The performance index to be minimized by $\vec{u}_e$ is following.

$$J = \int_{t_0}^{t_f}(\vec{X}_e^T Q\vec{X}_e + \vec{u}_e^T R\vec{u}_e)dt \tag{16}$$

where Q and R matrices are positive definite. The control $\vec{u}$ compensates the errors with respect to the desired position and velocity of the Optics cubesat, and the differences in orbital perturbations between the Optics and the Detector cubesats. When the differences in orbital perturbations are computed, they are easily inserted into Eq. (14). So only $\vec{u}_e$ should be evaluated by a control algorithm to make control $\vec{u}$ completed. A thruster system of the CANYVAL cubesats has a limitation on thrust magnitude. So, in the current study, the CMPC approach[7] is utilized to build an orbit controller to handle the limited control input. The error equation in Eq. (15) can be transformed to a discrete-time linear invariant system as the following form;

$$\begin{aligned}\vec{X}_e(t+1) &= A\vec{X}_e(t) + B\vec{u}_e(t) \\ \vec{y}(t) &= C\vec{X}_e(t)\end{aligned} \tag{17}$$

The constraint on control input can be imposed as

$$\vec{u}_{\min} \le \vec{u}_e \le \vec{u}_{\max} \tag{18}$$

$\vec{u}_e$ is computed by using the CMPC technique because constraint on control input should be satisfied. For the CMPC, the performance index in Eq. (16) can be equally transformed as follows [12-14]:

$$J' = \vec{X}_{e,N}^T P\vec{X}_{e,N} + \sum_{k=0}^{N-1}[\vec{X}_{e,k}^T Q\vec{X}_{e,k} + \vec{u}_{e,k}^T R\vec{u}_{e,k}] \tag{19}$$

P is the solution to the Eq. (20).

$$P = (A + BK_{LQR})^T P(A + BK_{LQR}) + K_{LQR}^T RK_{LQR} + Q \tag{20}$$

$$K_{LQR} = -(R + B^T PB)^{-1}B^T PA \tag{21}$$

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K_{LQR} is the algebraic Riccati equation of the unconstrained infinite horizon discrete-time Linear Quadratic Regulator (LQR) problem [7]. At time t+k, we have $\vec{X}_e(t+k) = A^k\vec{X}_e(t) + \sum_{j=0}^{k-1} A^j B\vec{u}_{e,t+k-1-j}$. This relationship is substituted into Eq. (19), in order to obtain a modified performance index as follows [7]:

$$J'' = \frac{1}{2}\vec{X}_e^T(t)Y\vec{X}_e(t) + \frac{1}{2}U^THU + \vec{X}_e^T(t)FU \quad \text{subject to} \quad GU \le E\vec{X}_e(t) + b \tag{22}$$

where $U \equiv [\vec{u}_e^T(t), \vec{u}_e^T(t+1), \ldots, \vec{u}_e^T(t+N)]^T$, $Y = 2(Q + T^TQ'T)$, $F = 2T^TQ'S$, $H = 2(R' + S^TQ'S)$

$$T = \begin{bmatrix} A \\ A^2 \\ \vdots \\ A^N \end{bmatrix}, Q' = \begin{bmatrix} Q & 0 & \ldots & 0 \\ 0 & Q & \ldots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \ldots & P \end{bmatrix}, S = \begin{bmatrix} B & 0 & \ldots & 0 \\ AB & B & \ldots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ A^{N-1}B & A^{N-2}B & \ldots & B \end{bmatrix}, R' = \begin{bmatrix} R & 0 & \ldots & 0 \\ 0 & R & \ldots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \ldots & R \end{bmatrix}$$

The U is the optimization vector at t, t+1, t+2…, $\vec{X}_e(t)$ is current error vector (i.e., states), and G, E, b matrixes can be easily obtained to satisfy the inequality constraints on control and states. The CMPC control law computes the optimal solution U to Eq. (22) at current time t, and applies to the system only the first control input from U, and repeats this process at every time step[7]. Equation (22) is a parametric Quadratic Program (QP) problem with inequality constraints. There are many techniques to solve QP problems. The QP problem described by Eq. (22) can be transformed into a dual QP problem because it is a convex QP. A dual QP, which is a simple and certifiable algorithm[15], is utilized in the current study. Then, the dual QP problem of Eq. (22) is obtained as follows[15]:

$$J''' = \frac{1}{2}y^TIy + (Dp + b)^Ty + d_p \quad \text{subject to} \quad y \ge 0 \tag{23}$$

where $I = GH^{-1}G^T$, $D = GH^{-1}F^T + E$, $d_p = \frac{1}{2}(F^Tp)^TH^{-1}(F^Tp) + p^TYp$. Here, p(t) is the current state (i.e., $\vec{X}_e(t)$) and y(t) is the optimization vector. If a solution y(t) to the dual QP problem in Eq. (23) is found, the optimal vector for original QP problem in Eq. (22) can be also obtained by using the following relationship[15]:

$$U(p) = -H^{-1}(G^Ty(p) + F^Tp) \tag{24}$$

A fast-gradient method[15,16] is used to solve the QP problem in Eq. (23). To prepare onboard operation, some parameters can be calculated prior to onboard computation as follows:

$$L = \sqrt{\sum_{i,j=1}^{m} |I_{i,j}|^2}, \quad H_G = H^{-1}G^T, G_L = G/L, H_F = H^{-1}F^T, E_L = E/L, b_L = b/L \tag{25}$$

For a given current state $p = \vec{X}_e(t)$ at current time t, some initializations are required for onboard computation.

$$g_p = H_Fp, p_D = -E_Lp - b_L, y_0 = y_{-1} = 0, z_{-1} = 0, \theta_0 = \theta_{-1} = 1 \tag{26}$$

For the onboard computation, the following iteration process is performed at every time t.

$$\begin{pmatrix} \theta_\tau = \frac{\sqrt{\theta_\tau^4 + 4\theta_\tau^2} - \theta_\tau^2}{2} \\ \beta_\tau = \theta_\tau(\theta_{\tau-1}^{-1} - 1) \\ \omega_\tau = y_\tau + \beta_\tau(y_\tau - y_{\tau-1}) \\ \hat{U}_\tau = -H_G\omega_\tau - g_p \\ U_\tau = (1-\theta_\tau)\hat{U}_{\tau-1} + \theta_\tau\hat{U}_\tau \\ y_{\tau+1} = \omega_\tau + G_L\hat{U}_\tau + p_D \end{pmatrix} \tag{27}$$

where τ is the iteration number. If no constraint is active, $y_{\tau+1} = 0$. There is termination criterion [15] for the iteration in Eqs. (27).

$$(GU_\tau - Ep - b) \le \varepsilon_g \text{ and } (J'' - J''') \le \varepsilon_\tau \tag{28}$$

where ε_g and ε_τ are stopping criterions. If the criterion in Eq. (28) is satisfied, the iteration process in Eqs. (27) is stopped and a optimization vector U is finally obtained at time t, and applies to the system only the first control input from U, and the process is repeated to compute a optimization vector U at next time step.

B. Formation Keeping Control

The shape of the original relative orbit for formation flying is changed after the astrometric alignment control because the control input affected the formation flying natural relative orbit. Formation keeping control is required to re-establish the original formation flying relationship between the Optics and the Detector cubesats. It is convenient for the cubesats to return their original formation after the astrometric alignment mode, because it is complex to design a new formation orbit in real time and there is no time enough to design a new optimal mission orbit, and going to a new optimal formation orbit may have a large cost. Actually, the original relative orbit designed

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for the VT formation flying is changed by orbital perturbations. The cost for returning to the unperturbed original relative orbit may take quite more than the cost for returning to the perturbed original relative orbit. Thus, in this study, the analysis is conducted for the cubesats to return to the perturbed original relative orbit, because the cost is less than the other cases. Let's define this process as formation keeping of the VT. A regulator problem is formulated for the Detector cubesat because the Detector should be returned to the state (position and velocity) of its perturbed original formation. Thus the perturbed original state will be a reference orbit, and its current state will be considered as relative error with respect its perturbed original state. Therefore we can use the Hill's equation described in Eq. (2) for the regulator problem. Similarly, a tracking problem is formulated for the Optics cubesat because the Optics should be returned to the state (position and velocity) of its perturbed original formation with respect to the Detector. The CMPC controller can be utilized as a controller for the regulator problem and the tracking problem. The associated state-space form for the Hill's equations is easily transformed into a discrete-time linear invariant system.

$$\vec{X}(t+1) = \begin{bmatrix} 0_{3\times3} & I_{3\times3} \\ A_1 & A_2 \end{bmatrix} \vec{X}(t) + \begin{bmatrix} 0_{3\times3} \\ I_{3\times3} \end{bmatrix} \vec{u}(t) \tag{29}$$

$$\vec{y}(t) = C\vec{X}(t)$$

$$A_1 = \begin{bmatrix} 3n^2 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -n^2 \end{bmatrix}, \; A_2 = \begin{bmatrix} 0 & 2n & 0 \\ -2n & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

where $\vec{u}=\begin{bmatrix} u_x & u_y & u_z \end{bmatrix}^T$, and n is the mean motion of the reference orbit. The state vector $\vec{X}=\begin{bmatrix} \Delta\vec{r}^T & \Delta\vec{v}^T \end{bmatrix}^T$ is difference between position and velocity of the corresponding cubesat and the position and velocity of the perturbed original formation. The constraint on control input can be imposed as $\vec{u}_{min} \le \vec{u} \le \vec{u}_{max}$. The optimization problem is to find the control, $\vec{u}$, that minimizes the performance index:

$$J=\frac{1}{2}\int_{t_0}^{t_f}\left(\vec{X}^T Q\vec{X}+\vec{u}^T R\vec{u}\right) dt$$

where Q is a real symmetric positive semi-definite matrix, and R is a real symmetric positive definite matrix. To compute $\vec{u}(t)$ at current time t, the process of the CMPC described in previous section can be applied. For the formation keeping case of the Optics cubesat, the state vector $\vec{X}=\begin{bmatrix} \Delta\vec{r}^T & \Delta\vec{v}^T \end{bmatrix}^T$ is difference between position and velocity of the Optics cubesat and the position and velocity of the perturbed original formation of the Optics cubesat. For the formation keeping case of the Detector cubesat, the state vector $\vec{X}=\begin{bmatrix} \Delta\vec{r}^T & \Delta\vec{v}^T \end{bmatrix}^T$ is difference between position and velocity of the Detector and the position and velocity of the perturbed original formation of the Detector.

C. Separate Control Strategy

The VTDM will adopt a separate control strategy to hold the desired astrometric alignment, because the cube satellites of the VTDM have low magnitude thrust capability. The Detector cubesat performs relative range control while the Optics cubesat conducts relative in-track and cross-track control with respect to the sun direction. Figure 4 explains the control directions for the two cubesats. To perform controls in proper directions, some coordinate transformations are necessary between the ECI (Earth-Centered Inertial coordinate) and a new $S_1S_2S_3$ coordinate. As shown in Figure 4, the $S_1S_2S_3$ coordinate is based on sun direction such that S_1 axis is the sun direction, and S_2 axis and S_3 axes are in perpendicular plane to the S_1 axis and satisfy a right-handed coordinate system. The control term ($\vec{u}_e$) due to the errors between a desired state and a current state of the Optics cubesat is calculated by using the CMPC technique. In this case, $\vec{X}_e$ is computed in the ECI coordinate using the Eq. (17), consequently the error control $\vec{u}_e$ is computed in the ECI coordinate. The error control $\vec{u}_e$ is transformed into the $S_1S_2S_3$ by using a rotation matrix between the ECI and the sun direction basis coordinate. The error control $\vec{u}_e$ described in the $S_1S_2S_3$ is assigned to each cubesat. The S_1 component of $\vec{u}_e$ is given to the Detector as the other two components of $\vec{u}_e$ are given to the Optics cubesat. Note that the sign of the S_1 component given to the Detector should be changed in opposite direction because the component is originally calculated for the Optics. The error controls assigned to the two cubesats are transformed into the ECI coordinate by using a rotation matrix between the $S_1S_2S_3$ and the ECI coordinate, proving a convenient method to apply Newton's 2nd law. The terms of orbital perturbations in Eq. (14) are also transformed with a similar rotation relationship. Now, the complete control law in Eq. (14) can be implemented for the two satellites.

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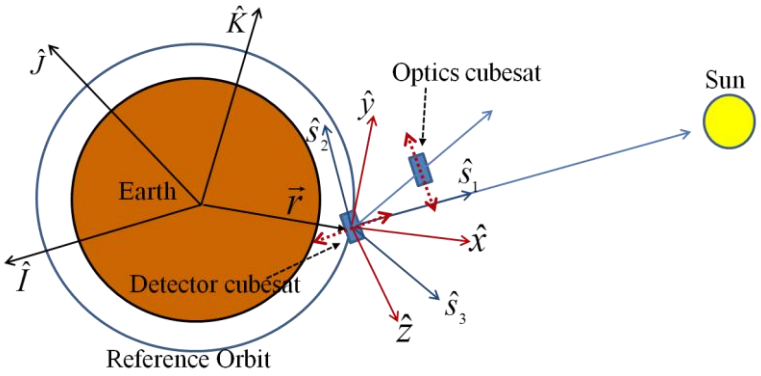


Figure 4. Definitions of coordinates for the ECI coordinate (i.e, IJK coordinate), xyz (the Hill coordinate) and $S_1S_2S_3$ (the sun direction basis coordinate).

V. Numerical Simulations

A. Orbital Perturbation Analysis

The dynamics of the Optics and Detector cubesats will vary from the theoretical two-body path due to perturbations caused by other celestial bodies (such as the Sun and Moon) and additional forces (such as non-spherical harmonics and atmospheric drag) not considered in Keplerian motion. The specific form of perturbation accelerations will depend on the number and type of perturbing sources. For the CANYVAL, we will examine perturbing effects of non-spherical harmonics, atmospheric drag, solar and lunar gravitations, and solar-radiation pressure [9]. Non-spherical gravitational acceleration has a great influence on the satellite, and the geopotential function ψ is expressed using non-spherical harmonic expansions in Eq. (30).

$$\psi = \frac{\mu_E}{r}\{1 - \sum_{n=2}^{N}\sum_{m=0}^{n}(\frac{R_E}{r})^n P_{nm}(\sin\phi)[C_{nm}\cos(m\lambda) + S_{nm}\sin(m\lambda)]\} \tag{30}$$

where, μ_E is the Earth's gravitational coefficient, R_E is the Earth's mean equatorial radius, r is distance of satellite from the Earth's center of mass, λ is longitude of satellite, Φ is geocentric latitude of satellite, P_{nm} is the associated Legendre functions, C_{nm} and S_{nm} are non-dimensional gravitational coefficients. The 70×70 order and degree of the coefficients are used in joint gravity model (JGM3) for this study. Perturbations by the solar gravity and the lunar gravity could be derived accurately although the celestial bodies are regarded as point masses. Therefore, the perturbing acceleration equation for the general n-body case is given by the Eq. (31).

$$\vec{a}_{nbody} = \sum_{i=1}^{n} GM_i\left[\frac{\vec{r}_{\oplus i}}{r_{\oplus i}^3} - \frac{\vec{r}_{sat,i}}{r_{sat,i}^3}\right] \tag{31}$$

In the equation, $\vec{r}_{\oplus i}$, $\vec{r}_{sat,i}$ mean the positions from the center of the Earth to the ith celestial body and satellite, respectively, and M_i is the mass of the ith celestial body. The atmospheric density model used in this study is exponential model; therefore, the atmospheric drag force has simple form of Eq. (32).

$$\vec{a}_D = -\frac{1}{2}(C_D A/m)\rho v_{rel}^2(\vec{v}_{rel}/|\vec{v}_{rel}|) \tag{32}$$

where A is the exposed cross-sectional area, m is the satellite mass and $\vec{v}_{rel}$ is the velocity vector relative to the atmosphere considering the constant the Earth's rotation without the effect of winds. Here, we assume the drag coefficient C_D for the two cubesats in the upper atmosphere is 2.3, cross-sectional area is 0.09 m^2 and the satellite mass 5 kg. In addition, atmospheric density model is

$$\rho = \rho_o \exp[-(h - h_0)/H]$$

where $\rho_o = 1.454 \times 10^{-13}$, $h_0 = 600\,km$, and $H = 71.835\,km$. Like drag, solar-radiation pressure is a non-conservative perturbation, but it becomes more pronounced at somewhat higher altitudes. The acceleration by the solar radiation pressure in Eq. (33) depends on the solar pressure P_{SR}, the reflectivity C_R, and the cross-sectional area to mass ratio.

$$a_{SRP} = -\frac{P_{SR}C_R A}{m}\left(\frac{\vec{r}_{sat\odot}}{r_{sat\odot}}\right) \tag{33}$$

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The solar pressure is $P_{SR} = 4.57 \times 10^{-6}\, N/m^2$, and the reflectivity C_R is assumed as 1.3, A is the exposed cross-sectional area (0.09 m^2), m is the cubesat mass (5 kg), $\vec{r}_{sat\odot}$ is position vector from satellite to the Sun. Moreover, the solar eclipse which is a key factor for the solar pressure is also considered depending on the simulation date and time.

Figure 5 shows the magnitudes of orbital perturbations on the Optics cubesat during 1.5 orbital periods of the satellite. The perturbation due to the non-spherical harmonics (70×70) is the largest one because it includes the J_2 term (order of $10^{-2}\mathrm{m/s^2}$). The other perturbations (lunar and solar gravitations, air drag, and solar radiation pressure) are less than $10^{-6}\mathrm{m/s^2}$ at about 600 km altitude. As shown in Figure 6, the orbital perturbations on the Detector cubesat are very similar to those of the Optics satellite because the two satellites are closely flying within about 40 m. Figure 6 indicates the "predicted" total perturbations and the "assumed true" perturbations for the two satellites. The "predicted" word means total perturbations summed and calculated from Eqs. (30), (31), (32) and (33). Since exact perturbation values cannot be obtained for a real mission, some random errors are added to the computed total perturbations in order to generate "assumed true" perturbations that are used for numerical simulations. The "assumed true" perturbations composed of non-spherical accelerations with maximum 0.1 % error, solar and lunar gravitational accelerations with maximum 10 % error, air drag with maximum 200 % error and solar radiation pressure with maximum 200 % error. The "predicted" total perturbations and the "assumed true" perturbations for the two satellites have the same order because J_2 term in non-spherical accelerations is dominated and other values are so small if compared with the J_2 term. Figure 7 shows differences in perturbations acted on the two cubesats. The differences in the "assumed true" perturbations between the two satellite used for numerical simulation in this study have order of $10^{-5}\mathrm{m/s^2}$, because the "predicted" total perturbations is in order of $10^{-2}\mathrm{m/s^2}$ and 0.1% error is randomly added to their values in order to generate the "assumed true" perturbations. The differences in the "predicted" total perturbations between the two cubesats also has order of $10^{-5}\mathrm{m/s^2}$. The differences in the perturbations due to non-spherical harmonics are exactly the same order as that of the "predicted" total perturbations, because the J_2 term (order of $10^{-2}\mathrm{m/s^2}$) in non-spherical harmonics is most dominant one among all of perturbations. The differences in other perturbations (lunar and solar gravitations, air drag and solar radiation pressure) are less than $10^{-10}\mathrm{m/s^2}$ at about 600 km altitude. At least, the differences in these other perturbations will not exceed their fundamental magnitude ($10^{-6}\mathrm{m/s^2}$) shown in Figure 5. Therefore, the differences in perturbations acted on the two cubesats of the CANYVAL mission will not significantly affect the relative motions of the satellite. Furthermore, the control strategy described in Eq. (14) can mitigate these differences in perturbations.

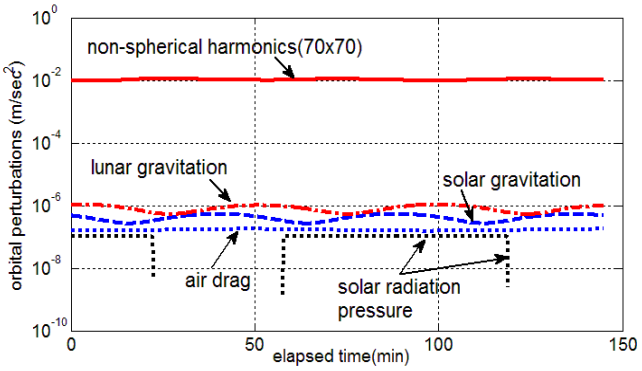


Figure 5. Magnitudes of orbital perturbations on the Optics cubesat.

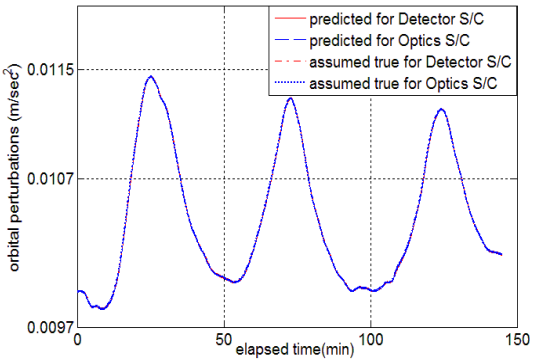


Figure 6. Magnitudes of total orbital perturbations and "assumed true" perturbations.

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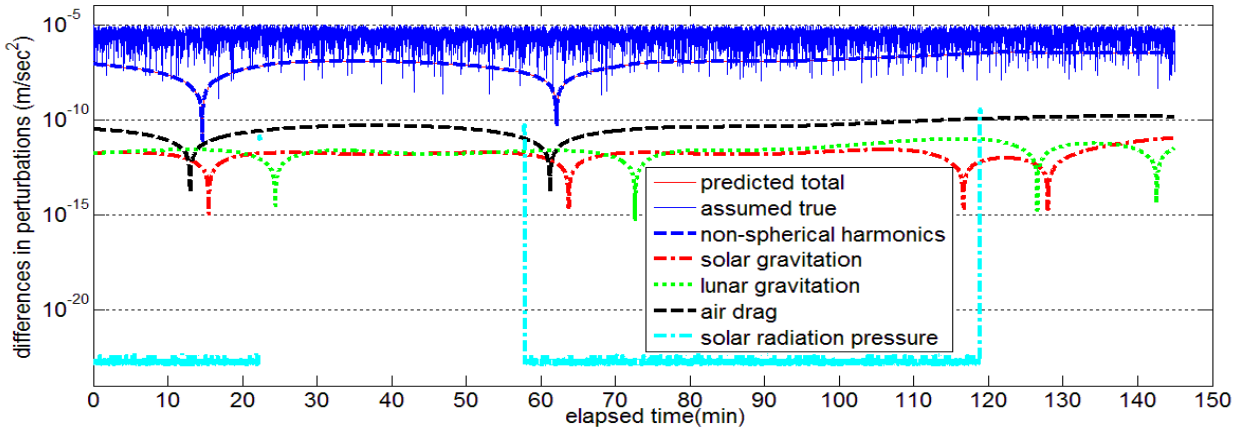


Figure 7. Differences in perturbations acted on the Optics and the Detector cubesats.

B. Controls for the Astrometric alignment and Formation Keeping

For the astrometric alignment control and the formation keeping control, the Q and R matrixes for the CMPC controllers are set as:

$$Q = \begin{bmatrix} {q_{11}}^{-2} I_3 & 0_3 \\ 0_3 & {q_{22}}^{-2} I_3 \end{bmatrix}, \qquad R = {R_1}^{-2} I_3$$

$$q_{11} = 3 \times 10^{-4}\ \text{km},\ q_{22} = 3 \times 10^{-6}\ \text{km/s},\ R_1 = 1 \times 10^{-7}\ \text{km/s}^2$$

The Q and R matrixes are chosen by ad-hoc method to yield an appropriate performance. The boundaries of control acceleration in the CMPC are set as:

$$\begin{pmatrix} \vec{u}_{\min} = -[0.2 \quad 0.2 \quad 0.2]^{\mathrm{T}} \text{mN/kg} \\ \vec{u}_{\max} = \ [0.2 \quad 0.2 \quad 0.2]^{\mathrm{T}} \text{mN/kg} \end{pmatrix} \tag{34}$$

The orbital perturbations described in the previous subsection are included for the simulations. The differences in the orbital perturbations between the two cubesats are fed back to the control strategy in Eq. (14).

The two cubesats are controlled to hold the astrometric alignment for 10 minutes from 79 minutes and 20 seconds elapsed time. The relative positions are well aligned with respect to the sun directions for approximately 5 minutes (during elapsed time of 85~91 minutes) as shown in Fig. 8 that plots the alignment error between the controlled Optics cubesat and the sun direction (i.e., the direction of line of sight from Detector cubesat to the Sun). The alignment error is less than 4.2 cm so that the desired astrometric alignment can be successfully conducted with mission requirements described in Table 2, for at least 5 minutes per orbit. After drift of 4 minutes followed by the astrometric alignment, formation keeping controls are conducted for 15 minutes. When the formation keeping maneuvers are finished, the keeping errors of the Detector and Optics cubesats with respect to their perturbed original formation are very small as shown in Figs. 9, 10 and 12. Figure 9 shows angle profiles of the controlled Optics cubesat with respect the sun direction. The angles are less than 0.12 deg during the inertial hold mode. The angles of the controlled relative orbit are the same as those from the perturbed original relative orbit after formation keeping maneuvers. Figure 10 plots relative distances between the controlled Optics and the Detector cubesat. Figure 11 enlarges the part of virtual telescope (astrometric) inertial alignment in Fig. 10. The relative distances are maintained within a range of 20.1 m ~ 20.5 m during the astrometric alignment. Figure 10 also confirms that the distances of the controlled relative orbit are the same as those from the perturbed original relative orbit after formation keeping maneuvers. Figure 12 shows formation flying shape of the controlled Optics cubesat with respect to the controlled Detector during both the inertial alignment mode and the formation keeping mode. Figure 13 informs Z-axis motion of the relative orbit in the Hill frame that is enforced by orbital perturbations counted in. The amounts in the Z-axis motion are not large enough to break down the desired astrometric alignment. Figure 12 also demonstrate that the two cubesats are well returned to the perturbed original relative orbit after the astrometric inertial hold mode, and the desired formation flying is well maintained in the presence of limited thruster magnitude. However, other numerical experiments denote that the formation keeping control (i.e., returning to the perturbed original relative orbit) will not be successful if the limitation on control magnitude is smaller than that in Eq. (34).

Figure 14 demonstrates the control profile obtained by the CMPC described. In this figure, the CMPC control profiles are divided into two parts; the first part in the plot is for the astrometric alignment control as the second part is for the formation keeping control. The first part control values of the CMPC controller are calculated in the Earth-

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Centered Inertial (ECI) coordinate because the controls are calculated using Eqs. (14) and (17) described in the ECI. The second part control values of the CMPC controller are calculated in the Hill's coordinate because the controls are calculated using Eq. (29) described in the LVLH frame. The control profiles obtained by the CMPC well satisfy the constraint on it for each axis in Eq. (34). By the reasons and strategy explained in subsection IV.C, the first part control values of the CMPC controller for the astrometric alignment are broken down into two portions in the sun-direction basis coordinate. One is for the Optics cubesat, the other is for the Detector. Based on the Newton's 2nd law, the control values should be applied in the ECI coordinate. So, the controls assigned to the two satellites are transformed into the ECI coordinate. The control values in the second part in Fig. 14 are also transformed into the ECI coordinate from the LVLH coordinate. Now, the whole controls can be conducted for the two cubesats, satisfying Newton's 2nd law. Figure 15 and 16 show the specific magnitude and the cumulative total delta V of the Optics cubesat, respectively. Figure 17 and 18 show the specific magnitude and the cumulative total delta V of the Detector cubesat, respectively. The total magnitudes of control values are bounded by 0.3464 mN ($=\sqrt{0.2^2+0.2^2+0.2^2}$) while each component is not bounded by 0.2 mN because the final control components in the ECI are transformed from those in other coordinate. Usually the control magnitude for the formation keeping maneuvers is bigger than that for the astrometric alignment maneuver. These values would be feasible amounts for the CANYVAL mission. The mission process for the astrometric alignment and formation keeping can be repeated may times as long as the fuel in each cubesat is available.

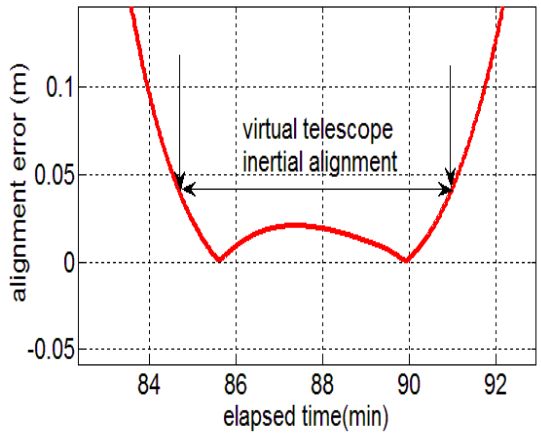


Figure 8. Alignment error between the two cubesats and the sun direction.

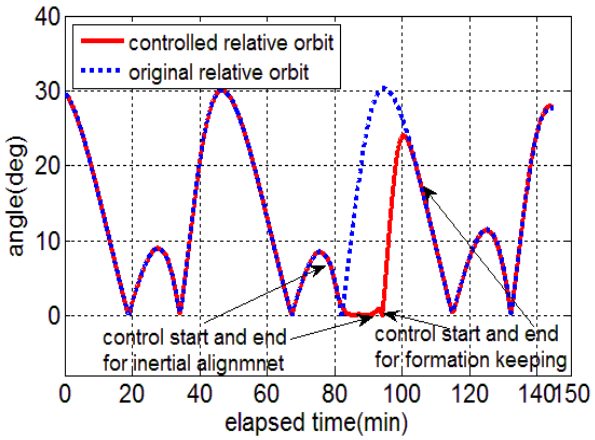


Figure 9. Angle profiles between the Optics cubesat and sun direction.

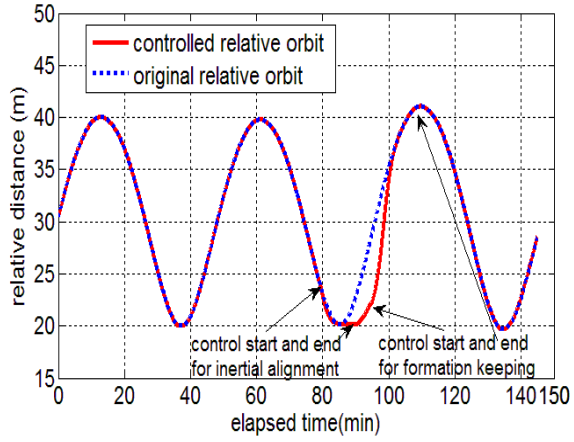


Figure 10. Relative distances between the Optics cubesat and the Detector cubesat.

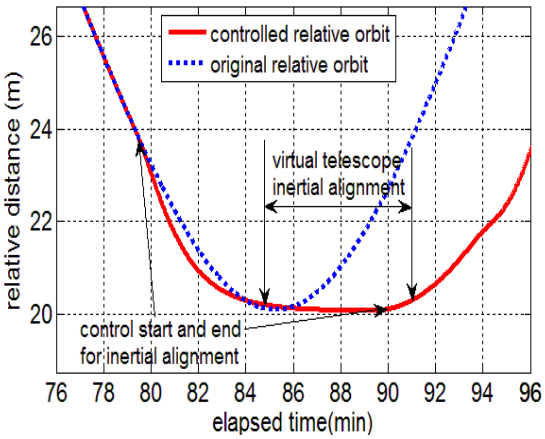


Figure 11. The plot enlarges the part of virtual Telescope (astrometric) inertial alignment

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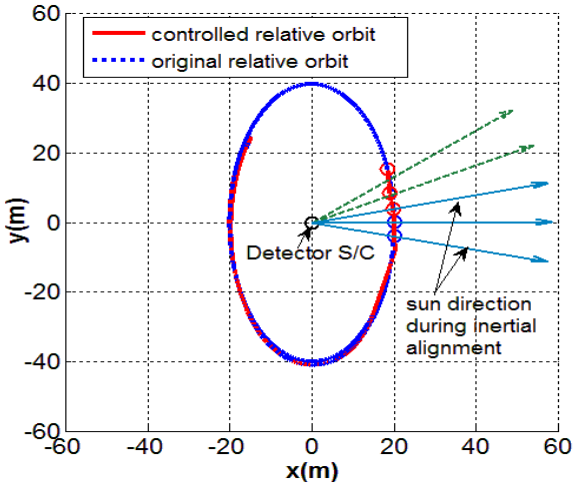


Figure 12. Relative orbit of the controlled Optics cubesat.

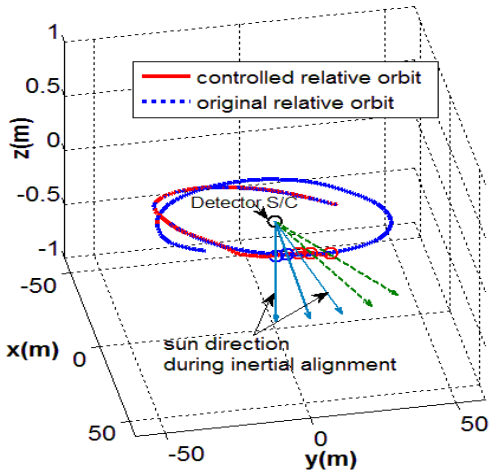


Figure 13. Z-axis component of the relative orbit.

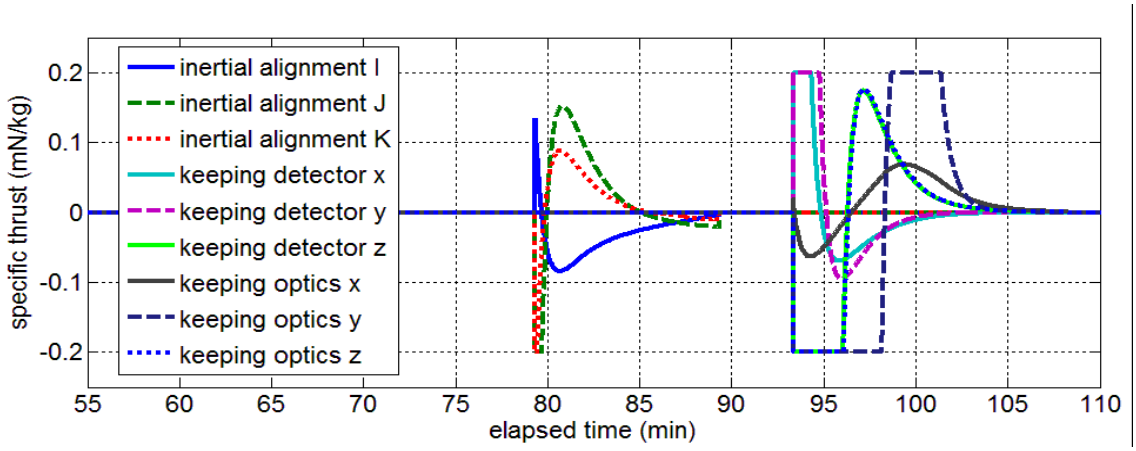


Figure 14. The CMPC control profiles for the astrometric alignment control (first part in the plot) and the formation keeping control (second part in the plot).

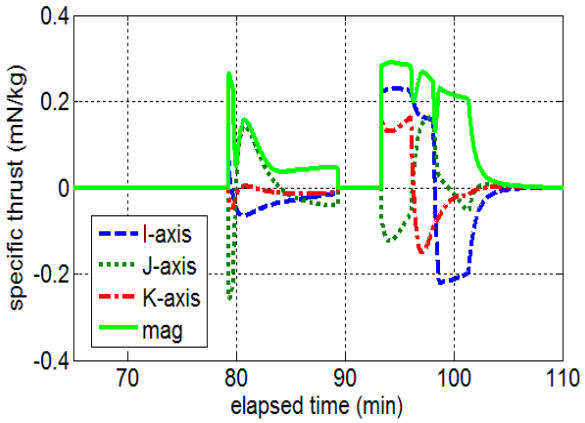


Figure 15. Specific thrust profile of the Optics cubesat.

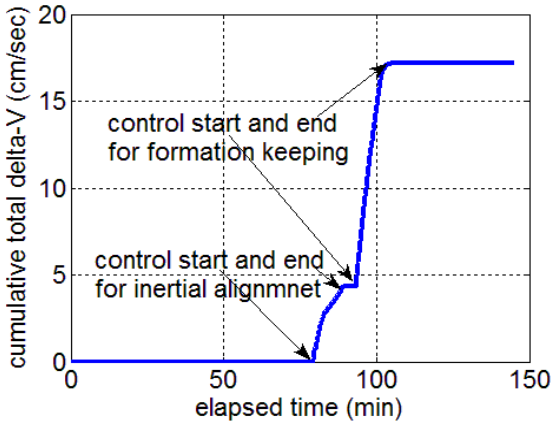
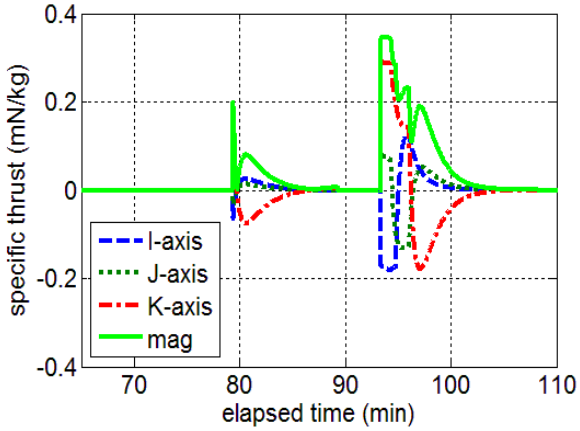


Figure 16. Cumulative total delta-V of the Optics cubesat.

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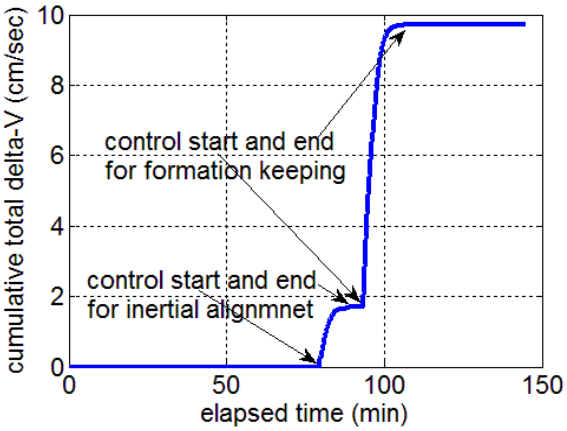


Figure 17. Specific thrust profile of the Detector cubesat. Figure 18. Cumulative total delta-V of the Detector cubesat.

VI. Conclusions

The relative dynamics for VT formation flying is designed using the Hill's equations and solutions and applied to a representative Virtual Telescope validation mission. A controller utilizing the Constrained Model Predictive Control (CMPC) technique is also developed to hold the astrometric alignment for a few minutes and to maintain the formation flying of the VTDM. Numerical simulations are performed for the astrometric alignment and formation keeping problems accommodating orbital perturbations. Analysis results confirm that the designed mission orbit is appropriate for the CANYVAL mission and the developed controller works properly for this representative VT mission. The numerical simulation results successfully demonstrate the capability of the proposed orbit design and control strategies by showing that the representative CANYVAL mission requirements are satisfied the during the astrometric alignment mode. Based on this work the flight dynamics and control for the CANYVAL Virtual Telescope validation mission is feasible even considering a very low thrust magnitude for orbit maneuvers.

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